



2014 Bored of Studies Trial Examinations

Mathematics Extension 2

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General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using black or blue pen. Black pen is preferred.
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- Show all necessary working in Questions 11 – 16.

Total Marks – 100

Section I Pages 1 – 5

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II Pages 6 – 18

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours 45 minutes for this section.

Total marks – 10

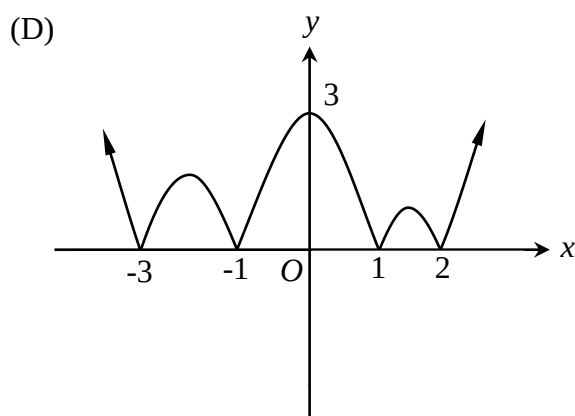
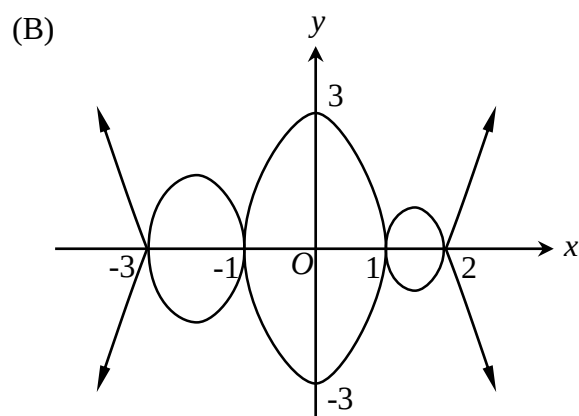
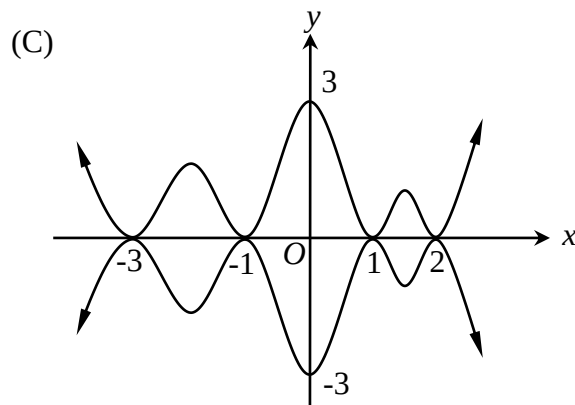
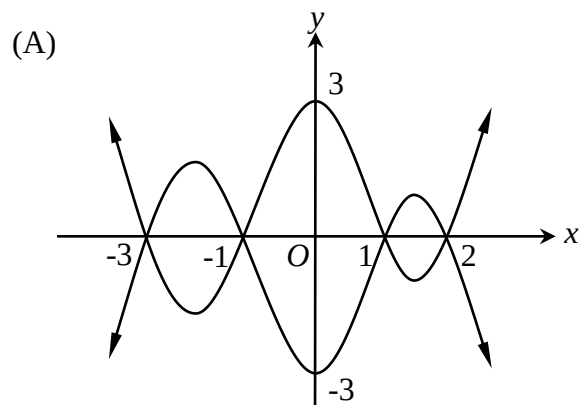
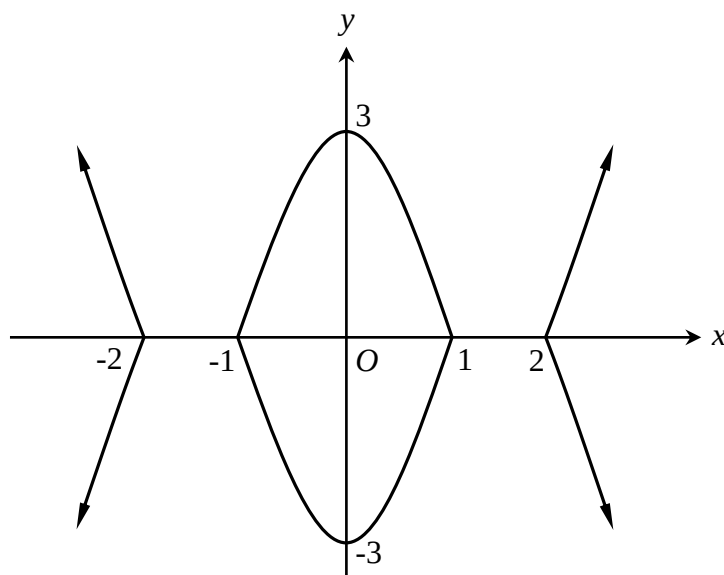
Attempt Questions 1 – 10

All questions are of equal value

Shade your answers in the appropriate box in the Multiple Choice answer sheet provided.

1 The following diagram shows the sketch of $|y| = f(|x|)$.

Which of the following curves is a possible sketch of $y^2 = [f(x)]^2$?

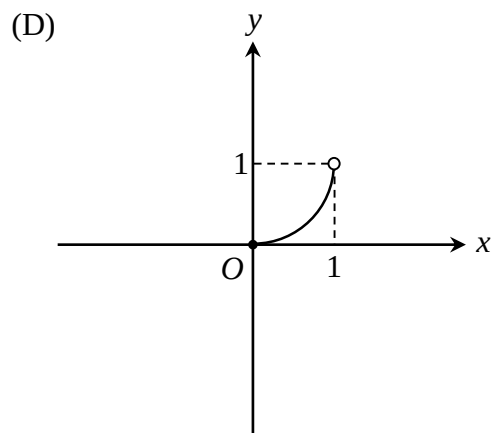
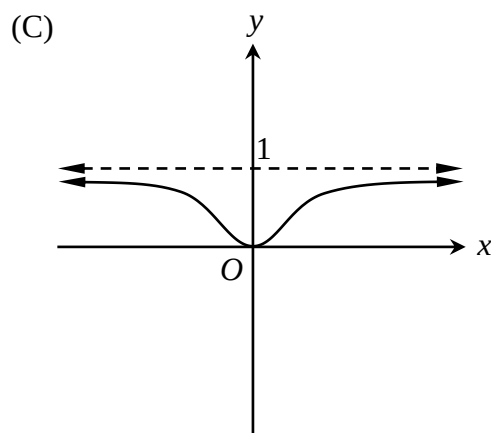
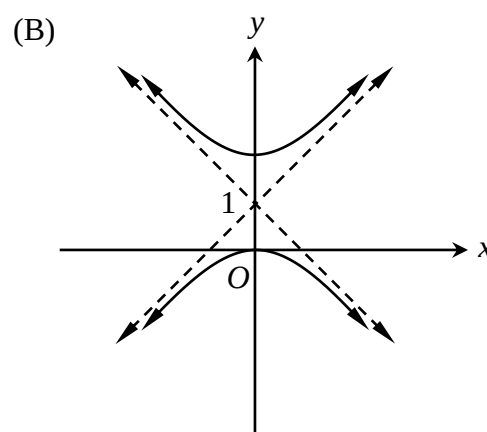
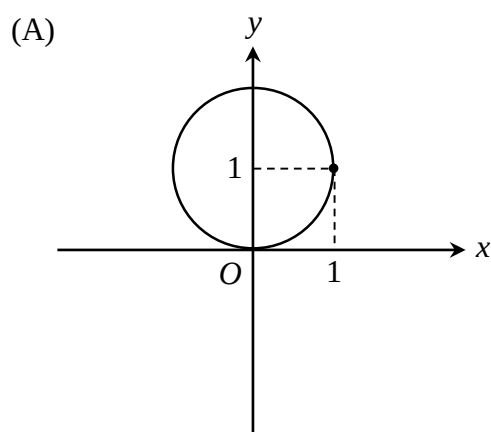


- 2 Consider an ellipse with the equation $\frac{x^2}{a_1^2} + \frac{y^2}{b_1^2} = 1$ and a hyperbola with the equation $\frac{x^2}{a_2^2} - \frac{y^2}{b_2^2} = 1$ for some positive constants a_1 , a_2 , b_1 and b_2 . The ellipse and hyperbola share the same foci. Which of the following statements is true?

- (A) The x -intercepts of the ellipse are greater than those of the hyperbola
 (B) The x -intercepts of the ellipse are less than those of the hyperbola
 (C) The ellipse and hyperbola share the same x -intercepts
 (D) The ellipse and hyperbola both do not have any x -intercepts

- 3 The implicit derivative of a curve is $\frac{dy}{dx} = \frac{x}{y-1}$.

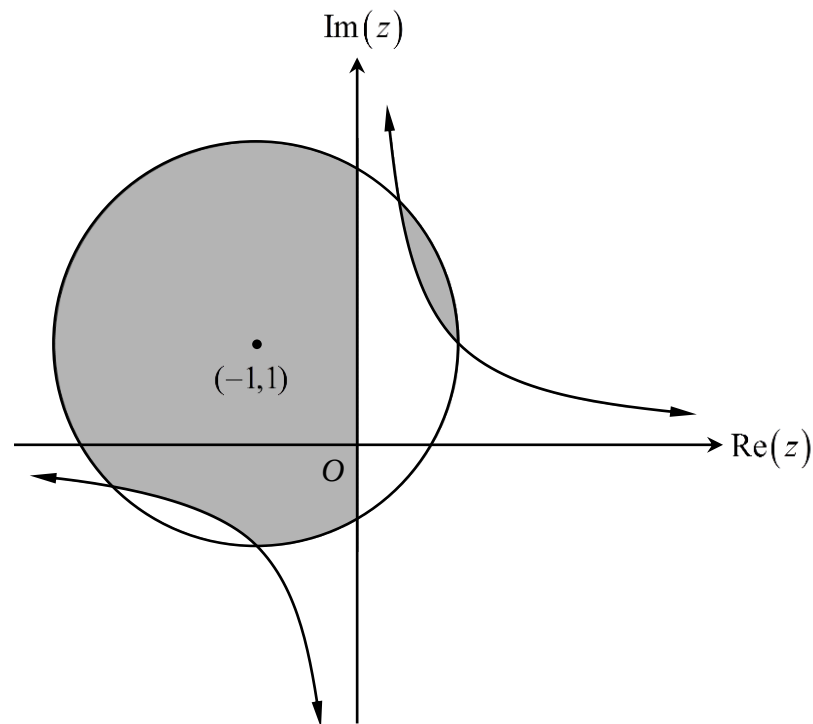
Which of the following sketches is a possible graph of the curve?



4 Suppose that $P(x)$ is a polynomial of degree n over the set of complex numbers where $n \neq 0$. What does the Fundamental Theorem of Algebra state about this polynomial?

- (A) $P(x)$ has at least one complex root
- (B) $P(x)$ has at least one real root
- (C) $P(x)$ has exactly n complex roots
- (D) $P(x)$ has exactly n real roots

5 Which of the following inequalities best define the region shown below?



- (A) $|z+1-i| \leq 2$ and $i \operatorname{Re}(z)(\bar{z}-z) \leq 2$
- (B) $|z+1-i| \leq 2$ and $i \operatorname{Re}(z)(\bar{z}-z) \geq 2$
- (C) $|z+1-i| \leq 2$ and $i(\bar{z}-z) \leq \frac{2}{\operatorname{Re}(z)}$
- (D) $|z+1-i| \leq 2$ and $i(\bar{z}-z) \geq \frac{2}{\operatorname{Re}(z)}$

6 Which of the following is the correct simplification of $\left(\frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta - i \cos \theta}\right)^4$?

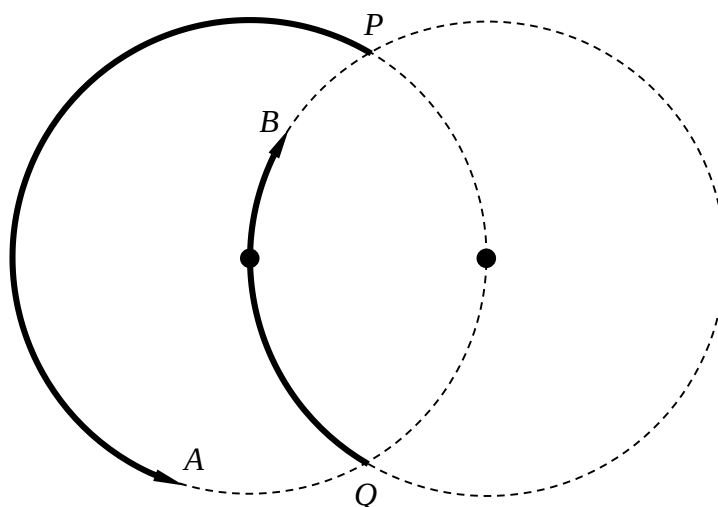
(A) $\sin 4\theta + i \cos 4\theta$

(B) $\sin 4\theta - i \cos 4\theta$

(C) $\cos 4\theta + i \sin 4\theta$

(D) $\cos 4\theta - i \sin 4\theta$

- 7 Two particles A and B travel along intersecting circles with equal radii, where the centre of one circle lies on the other circle as shown in the diagram below. Particle A is released from point P and moves anti-clockwise towards point Q along the major arc of one circle with constant angular velocity ω_A . Simultaneously, particle B is released from point Q and moves clockwise towards P along the minor arc of the other circle with constant angular velocity ω_B .



If the particles collide at point P , which of the following best describes the relationship between their angular velocities?

(A) $\omega_A = 2\omega_B$

(B) $\omega_B = 2\omega_A$

(C) $\omega_A = 3\omega_B$

(D) $\omega_B = 3\omega_A$

8 Let $P(x)$ be a non-zero polynomial with degree $n \geq 2$.

Which of the following statements is always true?

- (A) $P(x)$ has real quadratic factors if n is even.
- (B) $P(x)$ has at least one real root if n is odd.
- (C) $P(x)$ has only real roots if it is a real polynomial.
- (D) $P(x)$ has at least one real quadratic factor if it is a real polynomial.

9 Which of the following inequalities holds for all real x ?

- (A) $\sin x \leq x$
- (B) $\tan^{-1} x < e^x$
- (C) $2 \left| \sin \left(\frac{x}{2} \right) \right| < \sin x$
- (D) $|x| > \tan^{-1} x$

10 Let w be an n^{th} root of 1, where $w \neq 1$.

Which of the following is the exact value of

$$1 + 2(w + \bar{w}) + 3(w^2 + \bar{w}^2) + 4(w^3 + \bar{w}^3) + \dots + n(w^{n-1} + \bar{w}^{n-1})?$$

- (A) n
- (B) $-n$
- (C) $1 - n$
- (D) $-1 - n$

Total marks – 90

Attempt Questions 11 – 16

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Let $z = a + ib$ and $w = \frac{1}{a} + \frac{1}{b}i$, where $a > b > 0$.

(i) Calculate zw . **1**

(ii) Deduce that $\tan^{-1}\left(\frac{a}{b}\right) + \tan^{-1}\left(\frac{b}{a}\right) = \frac{\pi}{2}$. **1**

(b) Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{\sqrt{(1 + \cos x)(\sin x + \cos x)}}$. **3**

(c) Let $F(x)$ be a curve such that $F'(x) = \frac{x^3}{x^3 + 1}$.

(i) Sketch the curve $y = F'(x)$. **2**

(ii) Hence, sketch a possible graph of $y = F(x)$. **2**

(d) Consider the polynomial $P(x) = x^4 + ax^3 + bx + c$, where a , b and c are non-zero constants. **3**

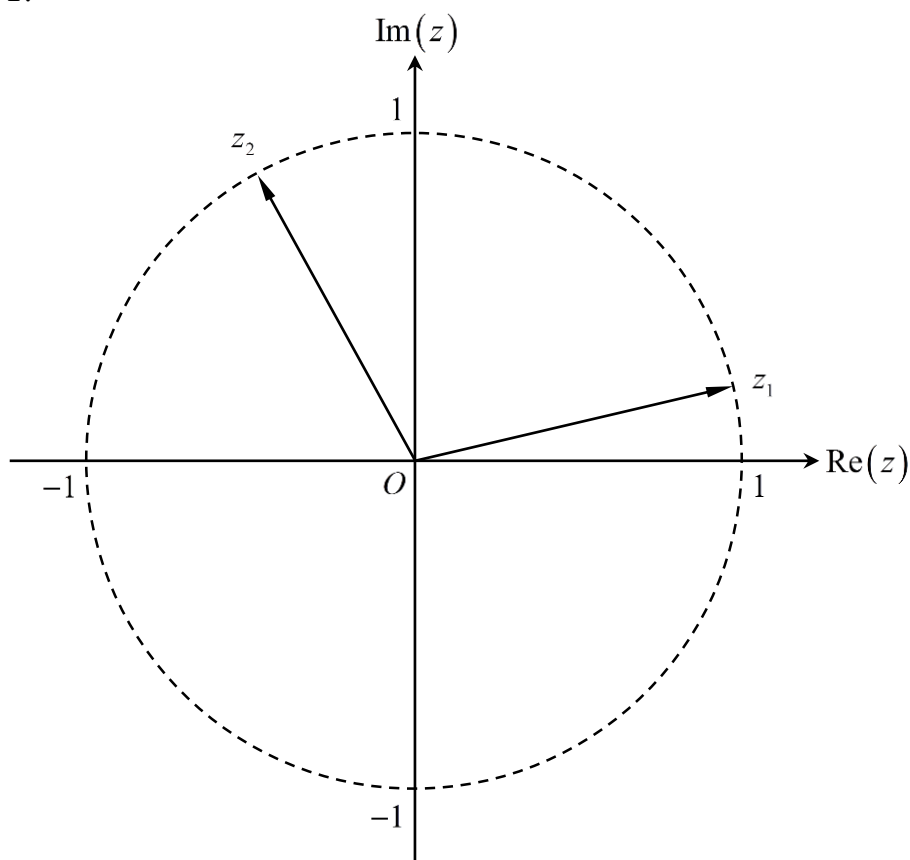
Prove that if $P(x)$ has a triple root, then $ab = 4c$.

(e) Find the equation of the locus of z if $\arg z = 2 \arg(z + 1)$. **3**

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) Let z_1 and z_2 be two non-zero complex numbers on the unit circle such that $z_1 z_2 \neq -1$.



By using vectors, or otherwise, prove that

3

$$\frac{z_1 + z_2}{1 + z_1 z_2}$$

is a real number.

- (b) The real polynomial $P(x) = x^3 + px + q$ has roots α, β and γ .

3

Find the polynomial with roots $\left(\frac{1}{\alpha} + \frac{1}{\beta}\right), \left(\frac{1}{\beta} + \frac{1}{\gamma}\right)$ and $\left(\frac{1}{\alpha} + \frac{1}{\gamma}\right)$

Question 12 continues on page 8

- (c) A section of flat road has a circular curve of radius r . A car moves along this curve and the tyres provide a frictional force F . The value of F is a maximum when the car travels with velocity V_0 . 4

Suppose the same road surface is now banked at some angle θ from the horizontal, where $0 < \theta < \frac{\pi}{2}$.

When the same car travels along the bank with velocity V_2 , it experiences no lateral forces. The value of the frictional force from the tyres is a maximum when the car travels with velocity V_1 .

Show that $\cos \theta = \frac{V_0^2}{V_1^2 - V_2^2}$.

- (d) Let z_1 , z_2 and z_3 be three complex numbers with modulus 1. 2

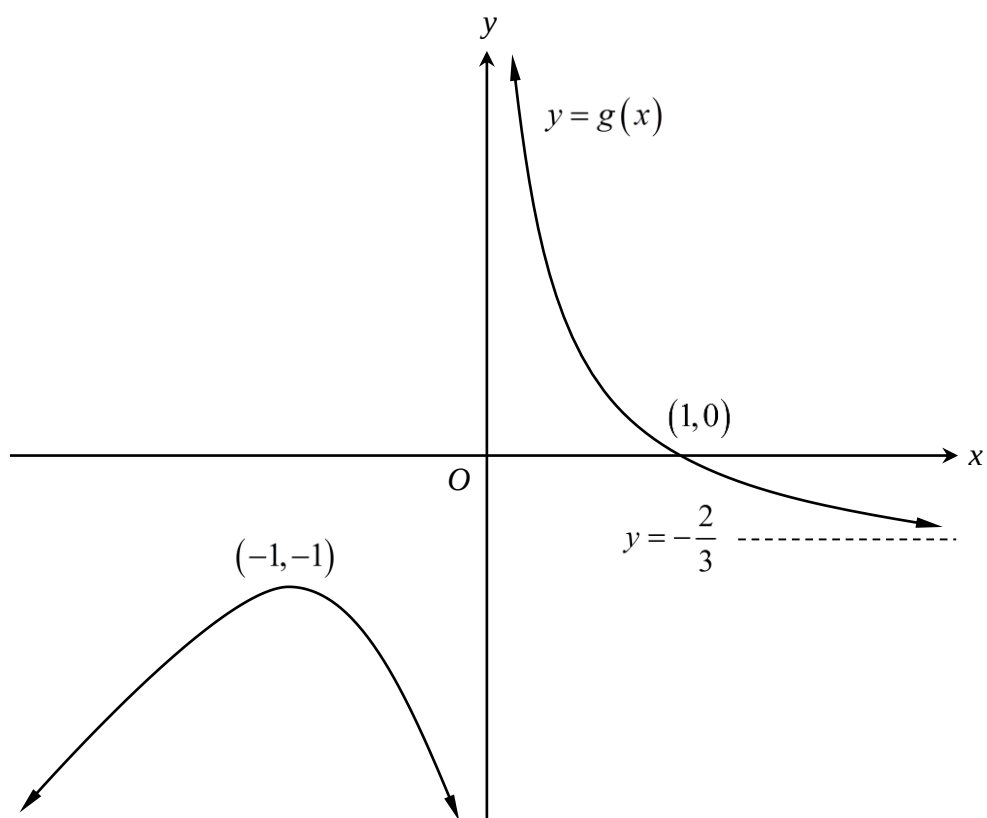
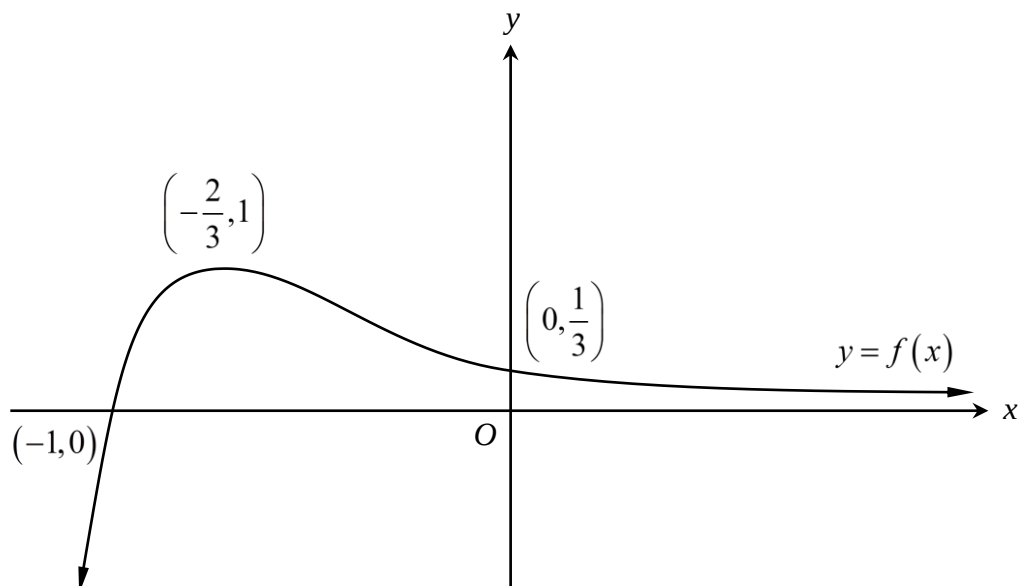
Prove that $|z_1 z_2 + z_1 z_3 + z_2 z_3| = |z_1 + z_2 + z_3|$.

Question 12 continues on page 9

- (e) The diagrams below show the graphs of $y = f(x)$ and $y = g(x)$.

3

Sketch a one-third page graph of $y = f(g(x))$.



End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) Let $P(z) = \sum_{k=0}^n a_k z^k$ be a monic non-constant polynomial with real coefficients, where $0 < |P(i)| < 1$.

- (i) Suppose $P(z)$ has p real roots. Let r_j be any real root. **1**

Show that $|r_j - i| \geq 1$, for any $j = 1, 2, 3, \dots, p$.

- (ii) Suppose $P(z)$ has q non-real roots c_k , where $k = 1, 2, \dots, q$. **2**

Prove that $|c_1 - i| \times |c_2 - i| \times \dots \times |c_q - i| < 1$

- (iii) Deduce that at least one non-real root α satisfies $|\alpha + i| < 1$. **2**

Question 13 continues on page 11

- (b) A particle A is dropped from a weather balloon. The equation of motion of particle A is

$$\ddot{x} = g - kv_A$$

where g is the acceleration due to gravity, k is a positive constant and v_A is the velocity of particle A .

T seconds later, an identical particle B is projected downwards from the same weather balloon with initial velocity $u \text{ ms}^{-1}$. The equation of motion of particle B is

$$\ddot{x} = g - kv_B$$

where v_B is the velocity of particle B .

- (i) Show that 4

$$T = -\frac{1}{k} \ln \left(\frac{g - ku}{g} \times \frac{g - kv_A}{g - kv_B} \right).$$

- (ii) Show that particle B 's displacement x_B is given by 3

$$x_B = \frac{1}{k} \left[u - v_B + \frac{g}{k} \ln \left(\frac{g - ku}{g - kv_B} \right) \right].$$

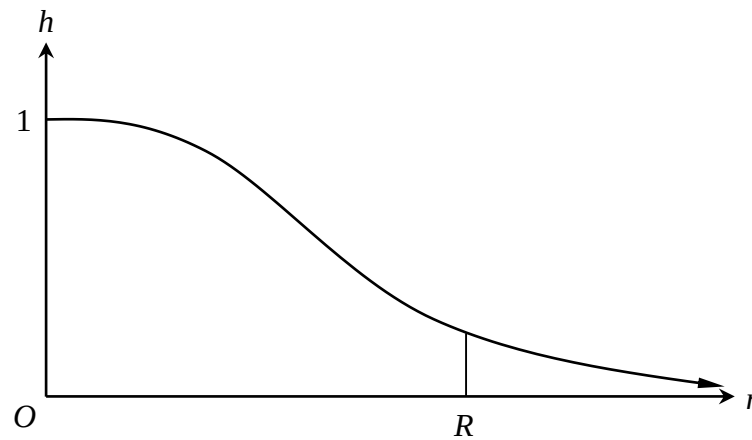
- (iii) Deduce that if particle B catches up with particle A , then particle B 3
must have been released no more than $\frac{u}{g}$ seconds after particle A .

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) The following diagram shows the curve $h = e^{-r^2}$, where $r \geq 0$.

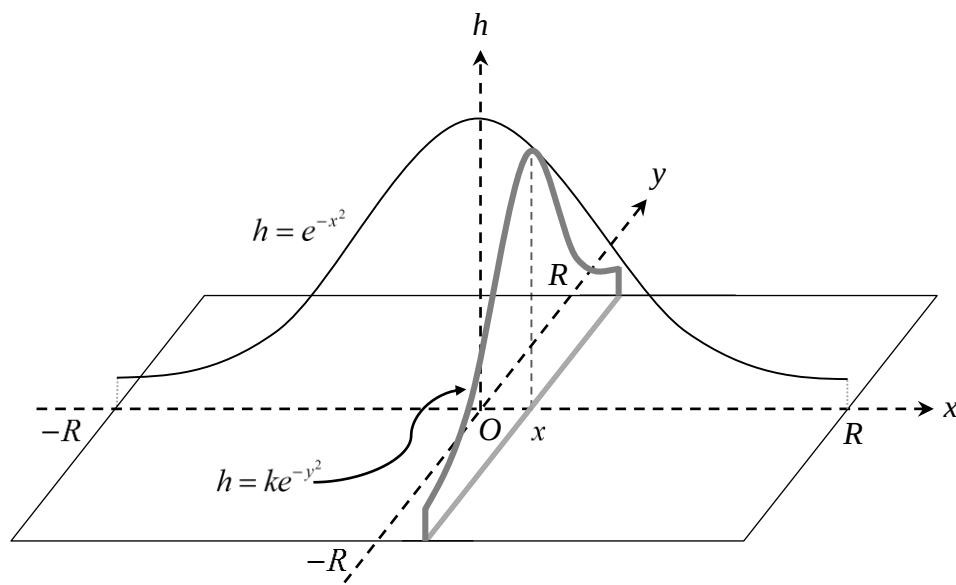
2



Use cylindrical shells to find the volume V_A of the solid formed when the region bounded by the curve, the coordinate axes and the line $r = R$ is rotated about the vertical axis.

Question 14 continues on page 13

- (b) The following diagram shows a square base on the xy – plane bounded by $-R \leq x \leq R$ and $-R \leq y \leq R$.



For a given value of x on the x axis, a thin slice is taken perpendicular to the square base and parallel to the y axis. The shape of the slice is determined by the equation

$h = ke^{-y^2}$, for some positive constant k , and is made such that its maximum point

lies on the curve $h = e^{-x^2}$ as shown in the diagram above. Let $I = \int_{-R}^R e^{-t^2} dt$.

- (i) Show that the area of this slice is given by $e^{-x^2} I$. 2
- (ii) Slices are taken continuously along the x axis and then added together, 2
forming a solid. Show that this solid has volume $V_B = I^2$.
- (iii) Consider the solid formed in part (a) with volume V_A . Suppose that 1
the rotated region is now bounded by the coordinate axis and the line $r = 2R$. Write down the volume V_C of the solid formed.
- (iv) Explain why $V_A < V_B < V_C$. 2
- (v) Deduce that $\lim_{R \rightarrow \infty} \int_{-R}^R e^{-t^2} dt = \sqrt{\pi}$. 1

Question 14 continues on page 14

(c)

- (i) Show that for $x \neq 1$ 2

$$1 + 2x + 3x^2 + \dots + nx^{n-1} = \frac{nx^{n+1} - (n+1)x^n + 1}{(x-1)^2}.$$

- (ii) A game is played where the probability of winning is p . 1

Let $P(k)$ represent the probability of winning for the first time on the k^{th} game.

Write down an expression for $P(k)$ in terms of p and k .

- (iii) Hence, prove that 2

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n kP(k) = \frac{1}{p}$$

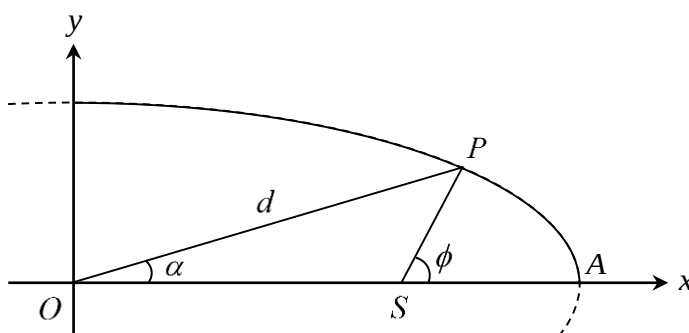
End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

(a) Evaluate $\int \frac{x^2}{(x \sin x + \cos x)(x \cos x - \sin x)} dx$. 4

(b) The diagram below shows a point P on the first quadrant of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $a > b$. Let A be the positive x -intercept of the ellipse.

Let d be the distance from P to the origin O , and let $\angle POS = \alpha$ and $\angle PSA = \phi$ where S is the focus of the ellipse.



(i) Show that $PS = \frac{a(1-e^2)}{1+e \cos \phi}$ 2

(ii) Show that $d \sin \alpha = \frac{a(1-e^2) \sin \phi}{1+e \cos \phi}$. 1

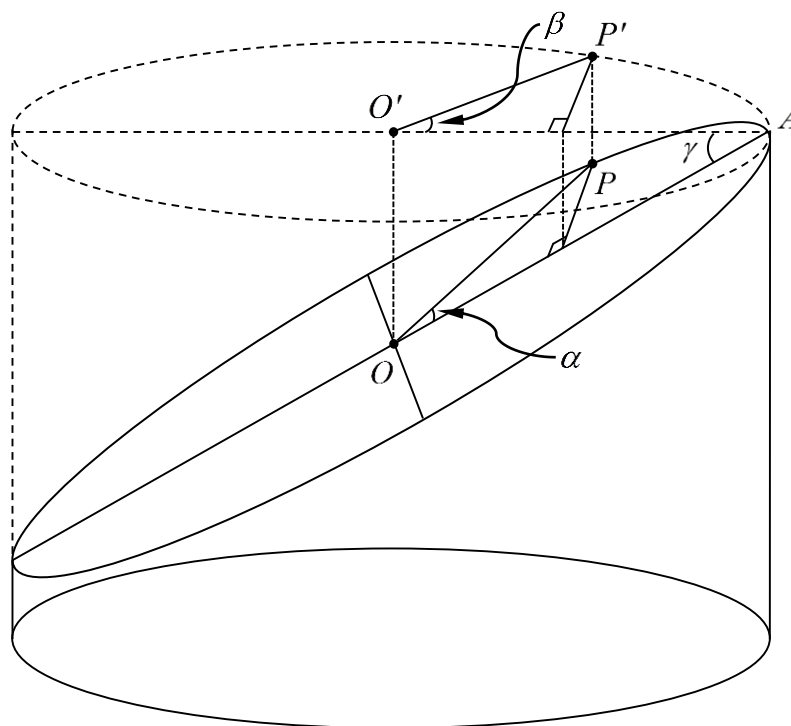
(iii) Show that $d \cos \alpha = \frac{a(e + \cos \phi)}{1+e \cos \phi}$. 1

(iv) Hence, prove that 3

$$d = \frac{a}{\sqrt{\frac{\sin^2 \alpha}{1-e^2} + \cos^2 \alpha}}.$$

Question 15 continues on page 16

- (c) The diagram below shows a cylinder that has been sliced at some angle γ from the horizontal, forming a solid with its top surface being an ellipse.



Consider a point P' on the edge of the top circular surface of the cylinder, which is directly above point P on the ellipse. The diameter of the top circular surface on the cylinder is directly above the major axis of the ellipse.

Let β be the angle subtended at the circle's centre by the minor arc AP' .

- (i) Using part (b), show that

2

$$\cos \beta = \frac{\cos \alpha}{\sqrt{\frac{\sin^2 \alpha}{1-e^2} + \cos^2 \alpha}}.$$

where e is the eccentricity of the ellipse at the top surface of the solid

- (ii) Hence, or otherwise, prove that $\tan \alpha = \tan \beta \cos \gamma$.

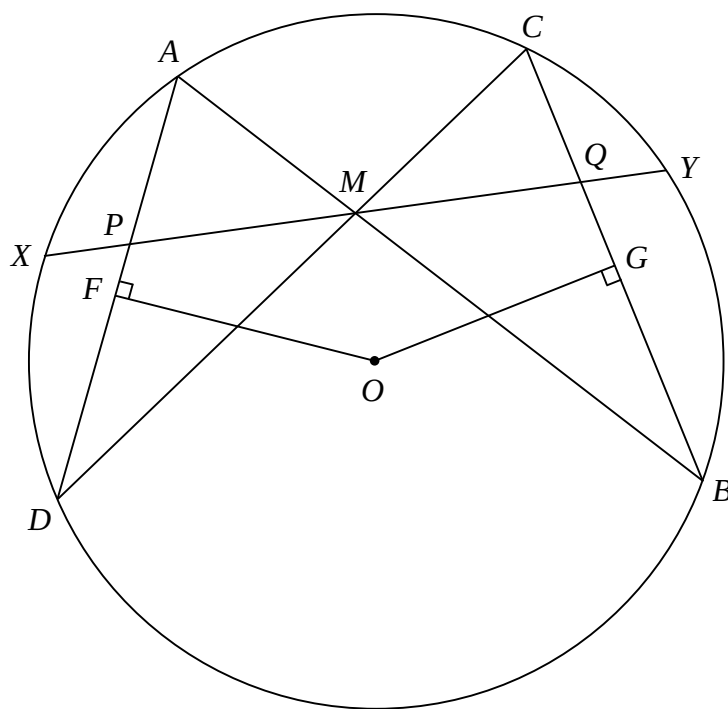
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End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) A chord XY with midpoint M is constructed in a circle with centre O . Two arbitrary chords AB and CD are drawn to pass through M . Let P and Q be the intersection points of AD and CB onto XY respectively.

Let F and G be the foot of the perpendiculars from O onto AD and CB respectively.



- (i) Prove that $\triangle FMA \parallel \triangle GMC$. 3
- (ii) Hence, or otherwise, prove that M is the midpoint of PQ . 3

Question 16 continues on page 18

(b) Let $I_n = (-1)^{n+1} \int_0^1 x^n e^x dx$, where n is a positive integer.

(i) Use mathematical induction to prove that

3

$$I_n = n! \left[1 - e \sum_{r=0}^n \frac{(-1)^r}{r!} \right]$$

(ii) Prove that $0 < \int_0^1 x^n e^x dx < \frac{e}{n+1}$.

1

(c) Consider a group of n people sitting in chairs labelled $1, 2, 3, \dots, n$ and a pile of n cards also labelled $1, 2, 3, \dots, n$. Each person picks a card randomly without returning it to the pile. A person has a matching card if the number on the card is the same as the number on their chair.

Let the number of ways where nobody receives a matching card be D_n , where $n \geq 2$.

(i) Explain why $D_n = (n-1)(D_{n-1} + D_{n-2})$.

1

(ii) Show that the expression

2

$$D_n = n! \times \sum_{r=0}^n \frac{(-1)^r}{r!}$$

satisfies the equation in (i).

(iii) Use part (b) to prove that when n is large, the probability of nobody picking their own card is approximately $\frac{1}{e}$.

2

End of Exam

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a \neq 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$